

Study Questions: Agents

1. What is interoception, and why does Seth (2013) argue it is fundamental to the sense of self?
2. Describe the role of the anterior insular cortex in interoceptive inference. What signals does it process?
3. How does the rubber hand illusion demonstrate that body ownership is an inference?
4. Explain how Active Inference accounts for the rubber hand illusion in terms of multisensory prediction error minimization.
5. What is depersonalization disorder? How does Active Inference explain it in terms of precision on interoceptive predictions?
6. What is body integrity identity disorder (BIID)? How does Active Inference explain the mismatch between the body and the body model?
7. How might the Active Inference framework account for phantom limb experiences?
8. What is alexithymia, and how does it relate to interoceptive inference?
9. How does the vagus nerve contribute to the brain's self-model? What role does the parasympathetic nervous system play?
10. Compare the Active Inference account of self-model with Damasio's somatic marker hypothesis. Are they compatible?
11. How does the brain distinguish between signals originating from inside the body and signals from the external environment?
12. What is the role of heartbeat-evoked potentials (HEP) in the neuroscience of self-awareness?
13. How does interoceptive sensitivity vary across individuals, and what cognitive consequences does this variation have?
14. How does the concept of allostasis (maintaining internal stability through change) relate to the self-model in Active Inference?
15. How might developmental changes in the self-model explain the gradual emergence of self-recognition in infants (mirror test)?
16. What neural evidence supports the claim that the anterior insula integrates interoceptive and exteroceptive signals?

17. How does Active Inference explain out-of-body experiences? What has gone wrong in the generative model?
18. Compare the Active Inference self-model with the Buddhist concept of *anattā* (no-self). Is the mapping appropriate?
19. How does emotional experience relate to interoceptive inference? Is an emotion a "prediction about interoceptive causes"?
20. Design a study that could test whether manipulating interoceptive precision (e.g., through breathing exercises) alters the strength of the rubber hand illusion.